Knoledge Distillation vs Lottery ticket Hypotesis

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Abstract

In this project, I compare two model compression techniques: Knowledge Distillation (KD) and the Lottery Ticket Hypothesis (LTH). I design controlled experiments on the MNIST1D dataset, which provides a non-trivial learning task while allowing fast training. Instead of focusing on a single compressed model, I train multiple student networks distilled from a larger teacher and identify multiple Lottery Ticket subnetworks, ensuring that all compressed models share the same parameter dimensionality. I compare their predictive performance, training time, and the structural and functional properties of their learned representations.

To gain deeper insight into the compressed solutions, I perform Representation Similarity Analysis using the Centered Kernel Alignment (CKA) metric, evaluate feature sparsity and effective dimensionality of the penultimate layer, and study the geometry of the loss landscape through Linear Mode Connectivity (LMC). These analyses are conducted both between dense and compressed models and across compressed models obtained with different compression strategies. The full implementation and experimental code are available at: https://github.com/Piras2024/Compressing-Networks-with-KD-vs-LTH.

A Introduction

In this project, I study model compression techniques with the goal of understanding how different strategies influence both the performance and the internal representations of neural networks. As modern deep learning models grow in size and computational requirements, compression methods are essential to reduce model complexity while maintaining predictive accuracy.

I focus on two widely used approaches: Knowledge Distillation (KD) [3] and the Lottery Ticket Hypothesis (LTH) [1]. Knowledge Distillation trains a smaller student network to mimic a larger teacher model, whereas the Lottery Ticket Hypothesis identifies sparse subnetworks within dense architectures that can be trained effectively on their own.

To compare these methods, I conduct experiments on the MNIST1D dataset [5], which allows extensive experimentation while preserving meaningful learning dynamics. I train multiple student networks via distillation and independently obtain multiple Lottery Ticket subnetworks, constraining all compressed models to have the same number of parameters. I then evaluate their predictive performance and training efficiency.

Beyond standard evaluation metrics, I analyze the learned representations using Centered Kernel Alignment (CKA) [4], assess the effective dimensionality and feature sparsity of the penultimate layer, and investigate Linear Mode Connectivity (LMC) [2] in parameter space. These analyses, performed both between dense and compressed models and across different compressed models, provide a detailed characterization of how KD and LTH influence representational capacity and functional similarity in compressed neural networks.

B Methods

B.1 Dataset

All experiments are conducted on the MNIST1D dataset [5], which provides a one-dimensional classification task that is sufficiently challenging to exhibit non-trivial learning behavior while allowing fast training and extensive experimentation. The input consists of signals of length 40 and the task is to classify them into 10 classes.

B.2 Model Architectures

Teacher Network. For Knowledge Distillation experiments, I use a highly overparameterized convolutional neural network (CNN) as teacher. The network consists of three convolutional blocks with channel sizes 64, 128, and 256, each followed by Batch Normalization, ReLU activations, and pooling operations. The final feature map is reduced to a single spatial dimension using adaptive average pooling. The classifier is composed of two large fully connected layers of sizes 1024 and 512, followed by the final 10-class output layer. The teacher network contains 999,562 trainable parameters.

Student Network. The student model is a significantly smaller CNN composed of two convolutional layers with 16 and 32 channels respectively, both using strided convolutions for downsampling. The representation is flattened and passed through a low-dimensional fully connected layer of size 19, followed by the output layer. The student network contains 8,059 parameters.

Dense Network for Lottery Ticket Experiments. To obtain Lottery Ticket subnetworks, I use a moderately overparameterized CNN composed of three convolutional blocks with channel sizes 32, 64, and 128, followed by a classifier with a hidden layer of size 64. This dense model contains 60,192 parameters.

B.3 Knowledge Distillation Protocol

I first train the teacher network for 500 epochs. Using the trained teacher, I then train multiple student networks through Knowledge Distillation [3], also for 500 epochs. In total, I train 10 students from the same teacher model.

To introduce variability across runs, I change the random seed used by the data loader generator, resulting in different batch orderings during training. All student networks share the same architecture and parameter dimensionality.

B.4 Lottery Ticket Hypothesis Protocol

Lottery Ticket subnetworks are obtained through Iterative Magnitude Pruning (IMP), following the Lottery Ticket Hypothesis framework [1]. Starting from the dense network, I perform 10 pruning rounds. At each round, 20% of the remaining parameters (by magnitude) are removed. After each pruning step, the pruned model is reinitialized and trained for 200 epochs. This procedure produces subnetworks (tickets) with approximately 8,078 parameters, matching the size of the student models.

Similarly to the distillation experiments, I obtain 10 different tickets by introducing randomness through the batch formation process, achieved by varying the data loader seed while keeping the same initial dense model initialization.

B.5 Matching Models

In addition to standard Lottery Tickets, I construct *matching models* to study the effect of partial weight rewinding. Instead of resetting weights to their initial values after pruning, the model is rewound to the weights obtained at epoch k of dense model training. In this project, I use $k = 5$. These models are trained following the same pruning schedule used for standard tickets.

B.6 Evaluation Metrics and Experimental Analyses

The comparison between Knowledge Distillation and Lottery Ticket compression methods is performed through multiple analyses.

- **Predictive performance:** test set accuracy and training time (minutes) are measured for each compressed model.
- **Representation similarity:** Centered Kernel Alignment (CKA) is computed on the penultimate feature layer to quantify similarity between compressed models and their corresponding reference models [4].
- **Linear Mode Connectivity (LMC):** the geometry of the solutions in parameter space is analyzed by testing LMC [2] between a ticket and the dense model, a matching model and the dense model, as well as across compressed solutions (pairs of students, tickets, and matching models).
- **Feature sparsity and effective feature dimensionality:** to study the functional capacity of compressed models:
  - Penultimate layer activations are collected using forward hooks during inference.
  - Feature sparsity is defined as the fraction of near-zero activations ($|h_i| < \epsilon$, with $\epsilon = 10^{-3}$) across the dataset.
  - Effective feature dimensionality is defined as the number of feature dimensions with variance above a small threshold (var $> 10^{-4}$).

These metrics allow a comprehensive comparison of structural and functional properties of compressed networks, providing insight into why LTH models often outperform KD models despite similar parameter budgets.

B.7 Linear Feature Connectivity via CKA

The formal definition of Layerwise Linear Feature Connectivity (LLFC) [6] requires that for every layer ℓ and every $\alpha \in [0, 1]$, there exists a scaling factor $c > 0$ such that:

$$c \cdot f^{(\ell)}(\alpha\theta_A + (1-\alpha)\theta_B) = \alpha f^{(\ell)}(\theta_A) + (1-\alpha) f^{(\ell)}(\theta_B) \tag{1}$$

In other words, the feature map of the interpolated model should equal the linear interpolation of the endpoint feature maps, up to a positive scaling factor.

Rather than testing this condition at every layer, I focus on the penultimate layer as a summary statistic, using CKA as the similarity measure. Since CKA is invariant to isotropic scaling, it naturally accounts for the factor c. Specifically, for each $\alpha \in \{0, 0.1, \ldots, 1.0\}$ I compute:

1. The weight-interpolated model $\theta(\alpha) = (1-\alpha)\theta_A + \alpha\theta_B$.
2. Its penultimate features $h_\alpha = f_{\theta(\alpha)}(X) \in \mathbb{R}^{N \times d}$.
3. The linearly interpolated features $h_{\text{lin}}(\alpha) = (1-\alpha)h_A + \alpha h_B$.
4. $\text{LLFC_CKA}(\alpha) = \text{CKA}(h_\alpha, h_{\text{lin}}(\alpha))$.

To quantify where failures originate, I also report an **LLFC error**: the test error obtained by passing $h_{\text{lin}}(\alpha)$ through $\theta(\alpha)$'s classifier head, bypassing the weight-interpolated feature extractor entirely. Since the head is identical in both the LMC and LLFC evaluations, any gap between LMC error and LLFC error directly reflects how much h_α diverges from $h_{\text{lin}}(\alpha)$ in a functionally relevant way.

I recompute BatchNorm running statistics on the training set after each weight interpolation to avoid BN miscalibration artifacts. I run the experiment on three model pairs: matching tickets (same initialisation), independent LTH tickets, and independent KD students.

C Results

I first compare Knowledge Distillation (KD) and the Lottery Ticket Hypothesis (LTH) as model compres-

Method	Test Acc. (%)	Time (min)
KD	96.68	2
LTH	98.89	12

Table 1: Performance comparison between compressed models.

Method	Mean CKA	Std
KD	0.82	0.028
LTH	0.95	0.005

Table 2: CKA similarity with dense reference models.

sion strategies in terms of predictive performance and computational cost. The results are reported in Table 1.

Lottery Tickets achieve higher final accuracy compared to distilled students. However, this improvement comes at a substantially higher computational cost due to repeated retraining during iterative magnitude pruning. In contrast, Knowledge Distillation allows obtaining compact models much more efficiently.

To better understand these performance differences, I analyze the similarity of the learned representations using the Centered Kernel Alignment (CKA) metric. In particular, I compute the similarity between compressed models and their dense reference models. The results are shown in Table 2.

Lottery Tickets learn representations that are highly similar to those of the dense model, whereas distilled students show lower similarity with the teacher. This observation is consistent with the higher accuracy obtained by Lottery Tickets.

I further investigate the geometry of the solutions through Linear Mode Connectivity (LMC). The analysis shows that LMC is not verified between Lottery Tickets and the dense model (Figure 1), while connectivity is observed when using matching models obtained through weight rewinding (Figure 2). This indicates that rewinding to early training weights helps maintain connectivity with the dense solution.

I then analyze representation similarity across independently trained compressed models. The results are summarized in Table 3.

Lottery Tickets exhibit higher mutual similarity than students, suggesting that iterative pruning tends to produce subnetworks that converge to more consistent representations across runs. Distilled students instead show higher variability in the learned representations.

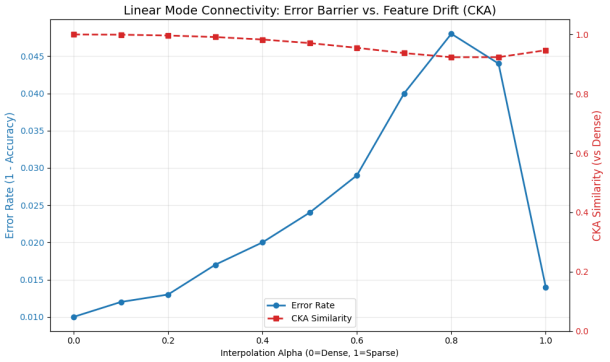

Figure 1: Connectivity between a Lottery Ticket and the dense model. No clear linear mode connectivity is observed.

Model	Mean CKA	Std
Students	0.879	0.031
Tickets	0.957	0.004

Table 3: CKA similarity across compressed models.

Finally, I evaluate Linear Mode Connectivity across compressed solutions. LMC is not verified between pairs of student models (Figure 3) nor between pairs of Lottery Tickets (Figure 4). However, connectivity is observed between matching models (Figure 5), further highlighting the role of early training dynamics.

C.1 Feature Sparsity and Effective Dimensionality

To better understand the differences in representational capacity between compressed models, I evaluate the penultimate layer activations in terms of feature sparsity and effective dimensionality. Feature sparsity is defined as the fraction of near-zero activations ($|h_i| < 10^{-3}$), and effective dimensionality is the number of feature dimensions with variance

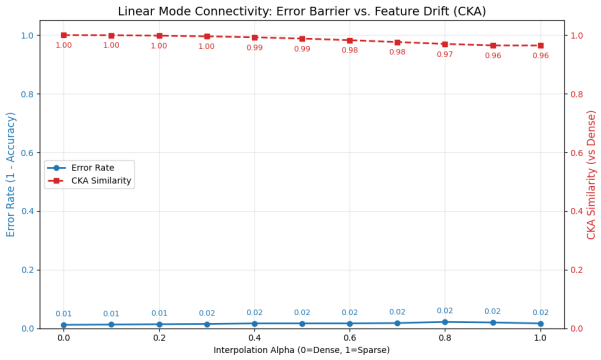

Figure 2: Connectivity between a matching model (rewound at epoch 5) and the dense model. Linear mode connectivity is restored.

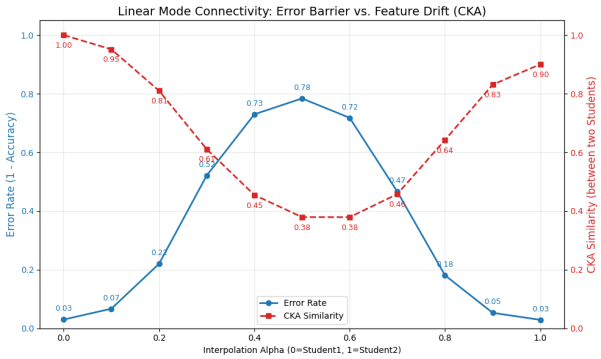

Figure 3: Connectivity between two student models. No clear LMC is observed.

above a small threshold (var $> 10^{-4}$). The results are reported in Table 4.

These results indicate that Lottery Tickets maintain almost full effective dimensionality in their penultimate layer, despite being highly parameter-sparse, whereas distilled students have a much smaller effective feature space due to the architectural bottleneck. This structural difference likely explains the higher CKA similarity of Lottery Tickets with dense models and their superior performance, even when Linear Mode Connectivity is not satisfied.

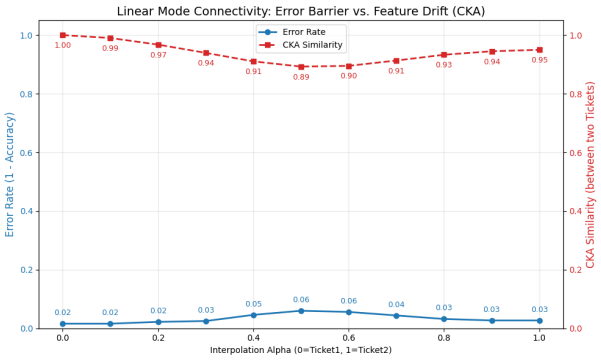

Figure 4: Connectivity between two Lottery Tickets. No clear LMC is observed.

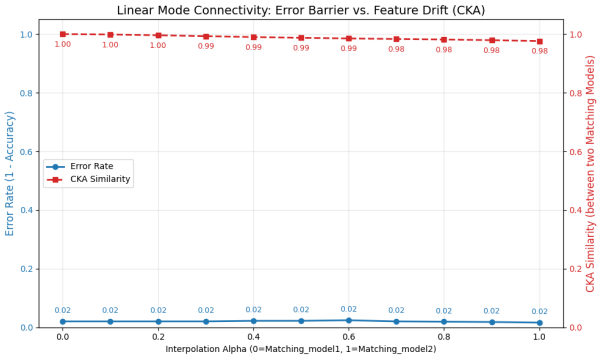

Figure 5: Connectivity between two matching models. Linear mode connectivity is observed.

C.2 Linear Feature Connectivity via CKA

Matching tickets satisfy both LMC and LLFC_CKA: the similarity curve stays flat near 1.0 and both error barriers are negligible, consistent with the expectation that a shared initialisation produces compatible feature spaces.

For LTH tickets, I observe a more nuanced result. LMC is not verified — the 3.1% error barrier at $\alpha = 0.5$ is consistent with earlier findings. Yet the LLFC barrier is exactly zero: replacing the weight-interpolated features h_α with the linearly interpolated features $h_{\text{lin}}(\alpha)$ and passing them through

Model	Feature Sparsity	Eff. Dim	Total Dim
Teacher	0.128	445	512
Student	0.089	16	19
Dense	0.0003	64	64
Ticket	0.0006	64	64

Table 4: Feature sparsity and effective dimensionality of the penultimate layer for all models.

Pair	LMC	LLFC	$\overline{\text{CKA}}$	CKA@0.5
Matching	0.006	0.003	0.997	0.995
LTH (indep.)	0.031	**0.000**	0.980	0.950
KD (indep.)	0.749	**0.162**	0.772	0.407

Table 5: LMC barrier, LLFC barrier, and LLFC_CKA at the penultimate layer.

the same head W_α recovers near-perfect accuracy at every α. The LLFC_CKA drops only mildly to 0.95 at $\alpha = 0.5$. While this does not constitute a formal verification of LLFC (which would require testing at every layer), it suggests that the penultimate representations of the two tickets are largely compatible in a linear sense. The LMC barrier therefore appears to originate in the weight interpolation of the feature extractor, which fails to reproduce the linear mixture of endpoint representations.

For independent KD students, I find a large LMC barrier (74.9%) and a significant LLFC barrier (16.2%). The LLFC_CKA collapses to 0.41 at $\alpha = 0.5$, indicating genuine feature incompatibility between the two models.

D Discussion

The experimental results provide several insights into the behavior of Knowledge Distillation (KD) and the Lottery Ticket Hypothesis (LTH) when used as model compression strategies.

First, Lottery Tickets consistently achieve higher test accuracy than distilled student models. The representation similarity analysis further shows that tickets learn internal representations that are significantly closer to those of the dense reference model, as indicated by the higher CKA scores. This suggests that preserving representational structure is

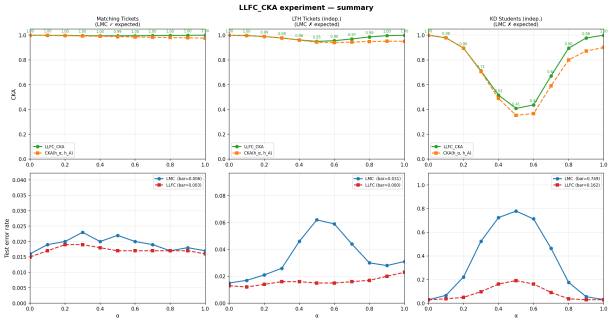

Figure 6: LLFC_CKA and error barriers across interpolation α for all three model pairs.

an important factor in achieving strong performance in compressed models. Interestingly, this improved alignment is observed even though Linear Mode Connectivity (LMC) is not verified between standard Lottery Tickets and the dense model.

Connectivity with the dense solution can instead be restored when using matching models obtained through weight rewinding. This indicates that early training dynamics play a critical role in shaping the geometry of the solution space. Exploring alternative rewinding strategies or hybrid compression procedures that combine iterative pruning with partial training reuse may therefore represent an interesting direction for future work.

The LLFC analysis adds a layer of detail to the LMC results. For matching tickets and KD students, the findings are consistent with expectations: matching tickets exhibit low barriers under both metrics, while students show high barriers under both, in line with their higher representational variability already reported. For LTH tickets, the cross-model CKA analysis already established that the two models produce similar representations. What the LLFC experiment adds is the observation that this similarity is sufficient for classification: passing the linearly interpolated features through the interpolated head recovers near-perfect accuracy at every α, with zero LLFC barrier. This localises the LMC barrier to the weight interpolation of the feature extractor, which fails to reproduce the linear mixture of endpoint representations, rather than to any fundamental incompatibility

between the learned feature spaces.

Another important aspect that emerges from the experiments is the difference in architectural flexibility between the two compression approaches. In the Knowledge Distillation setting, the student architecture is fixed and constrained to contain a very low-dimensional penultimate representation layer of size 19, which is substantially smaller than the 512-dimensional representation used by the teacher network. This architectural bottleneck limits the capacity of the student to reproduce the internal structure of the teacher's solution.

In contrast, the Lottery Ticket procedure allows the pruning process to adaptively determine where model capacity should be preserved. An analysis of representative tickets shows strong *parameter sparsity* (around 86% globally), but almost no *feature sparsity*. In particular, while the dense and ticket models both operate in a 64-dimensional penultimate feature space, the effective dimensionality remains equal to the full dimension (64/64 active features), indicating that pruning removes weights without collapsing the representation space.

This behavior is fundamentally different from that of the distilled student, where the combination of architectural compression and training dynamics leads to a reduced effective dimensionality (about 16/19 active features). The teacher model itself exhibits moderate natural sparsity due to ReLU nonlinearities (approximately 13% inactive features out of 512), but still maintains a very rich representation space.

These observations suggest that Lottery Tickets are **structurally sparse but representationally dense**. The ability of iterative magnitude pruning to preserve full feature dimensionality while redistributing capacity across layers provides a natural advantage over fixed-architecture distillation. This helps explain both the higher CKA similarity with dense models and the improved test accuracy obtained by tickets.

Overall, the results indicate that maintaining a sufficiently expressive feature space is more important than simply reducing parameter count. Methods such as LTH that allow adaptive capacity allocation may therefore produce compressed models that remain closer to dense solutions in representation space, even when direct connectivity in parameter space is not preserved.

References

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